

Dataset	Method	Nodes	GPU time (s)	$C_{\max}$	Δ_{GPU}	Pass
GSE24080	Exact degree	27,338	1.172 ± 0.035	0	—	—
	$\varepsilon = 10^{-1}$	44–53	0.090 ± 0.020	7.69×10^{-2}	3.40×10^{-4}	5/5
	$\varepsilon = 10^{-2}$	45–54	0.085 ± 0.007	7.09×10^{-3}	3.71×10^{-4}	5/5
	$\varepsilon = 10^{-3}$	46–54	0.086 ± 0.006	8.82×10^{-4}	1.31×10^{-3}	4/5
TCGA LGG	Exact degree	198,033	66.377 ± 1.196	0	—	—
	$\varepsilon = 10^{-1}$	37–49	0.320 ± 0.028	7.01×10^{-2}	4.47×10^{-3}	5/5
	$\varepsilon = 10^{-2}$	38–50	0.312 ± 0.006	6.51×10^{-3}	3.80×10^{-3}	5/5
	$\varepsilon = 10^{-3}$	39–51	0.310 ± 0.011	5.80×10^{-4}	5.95×10^{-3}	2/5

Table 1: Cox explanations on an Apple M4 Pro GPU using JAX-Metal FP32, 30 fixed training-background samples, and five distinct held-out patients. Time is mean $\pm$ sample standard deviation across patients, with one synchronized call per patient and method; node ranges span patients. Approximate results are newly measured; exact-degree rows reuse the matching previously measured baseline with identical model, inputs, numerical sources, software versions, and memory plan. $C_{\max}$ is the largest analytical absolute quadrature-error bound across patients. Δ_{GPU} is the largest absolute difference from the matching GPU degree-exact output over all patients and features. Pass counts patients whose maximum feature-wise difference is at most ε . The certificate excludes floating-point error; GPU degree-exact output is itself subject to rounding. Warm-up and rule construction are excluded; automatic node selection is included. Previously measured CPU construction of the cached exact rules took 10.97 s (GSE24080) and 561.92 s (TCGA LGG); fresh approximate-rule construction took 0.386–0.563 ms (range of medians over five constructions).

1 High-dimensional experiments with QuadraSHAP

To assess whether Shapley attribution remains practical when the number of features greatly exceeds the number of patients, we consider two genomic survival datasets: GSE24080 multiple-myeloma gene expression [1] and TCGA lower-grade glioma (LGG) DNA methylation [2]. Genome-wide assays measure many molecular variables per patient, whereas the cohorts with matched survival follow-up contain only hundreds of patients. After eligibility filtering, GSE24080 contains 553 patients and 54,675 expression probe sets, and TCGA LGG contains 511 patients and 396,065 methylation sites. These are probe- or site-level features, retained without gene aggregation. We use the supplied 339/214 training/validation split for GSE24080 and a 383/128 split stratified by event status for TCGA LGG (seed 42), giving an extreme small- n , large- d setting.

We model overall survival using a Cox proportional-hazards model, which incorporates right-censored observations: patients whose death has not been observed by the end of follow-up. Its relative hazard factors across features, making QuadraSHAP applicable directly:

$$h(t \mid x) = h_0(t)f(x), \quad f(x) = \exp(\beta^\top x) = \prod_{j=1}^d \exp(\beta_j x_j).$$

To stabilize estimation when $d \gg n$, we shrink the Cox coefficients towards zero using a ridge penalty, with the implementation’s penalty parameter fixed a priori at $\alpha = 0.01d$. We use the Breslow approximation for patients with equal recorded event times. Missing values are imputed using training medians, and centering and scaling use training statistics. All retained features have nonzero fitted coefficients.

We explain the relative hazard using 30 training-background patients, sampled without replacement with seed 42 and held fixed across methods. Specifically, absent features are replaced by the

corresponding entries of each background row and predictions are averaged:

$$v_x(S) = \frac{1}{30} \sum_{b=1}^{30} f(x_S, z_{\overline{S}}^{(b)}).$$

The resulting game is a weighted sum of product games. We explain the first five held-out patients in each saved split with the degree-exact rule $m_q = \lceil d/2 \rceil$ and with patient-specific node counts selected to certify absolute quadrature tolerances $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-3}\}$. The timing experiments use JAX-Metal FP32 on an Apple M4 Pro, an 8 GiB memory-planning budget, and one background row per block. We record one synchronized explanation call per patient and method after warming up each distinct dataset/node-count configuration. Reported times include automatic node selection, data transfers, and host accumulation, but exclude model fitting, warm-up, and cached quadrature-rule construction.

The approximate GPU experiments were rerun for these three tolerances. The degree-exact baseline is reused from the matching five-patient experiment; the model, background rows, patient inputs, numerical source hashes, software versions, and GPU memory plan were verified to match.

Table 1 shows that degree-exact quadrature remains manageable even at these dimensions: 27,338 and 198,033 nodes require 1.17 s and 66.38 s per patient on average, respectively. For latency-sensitive use, the certified node budgets reduce the measured latencies to 85–90 ms for GSE24080 and 310–320 ms for TCGA LGG. These subsecond latencies support interactive, on-demand Shapley explanations once the model and quadrature rules are ready. At $\varepsilon = 10^{-1}$, the speedups relative to the reused exact means are $13.1\times$ and $207.1\times$, respectively. The exact-rule construction costs are substantial but can be amortized across patients by caching the nodes and weights.

The analytical certificate and measured numerical discrepancy answer different questions. For patient p , let ϕ_p be the exact Shapley vector for the fixed background game and $\phi_{m_p,p}$ its quadrature approximation in real arithmetic. The selected node count satisfies

$$\|\phi_{m_p,p} - \phi_p\|_\infty \leq C_p \leq \varepsilon, \quad C_{\max} = \max_{p=1,\dots,5} C_p.$$

Thus $C_{\max}$ is an evaluated analytical bound, not an observed error. The table’s empirical discrepancy instead compares two floating-point outputs:

$$\Delta_{\text{GPU}} = \max_p \|\hat{\phi}_{m_p,p}^{\text{GPU}} - \hat{\phi}_{\text{exact},p}^{\text{GPU}}\|_\infty.$$

Every selected rule satisfies its analytical quadrature bound. At $\varepsilon = 10^{-1}$, the observed GPU discrepancy is within the requested threshold for 5/5 GSE24080 and 5/5 TCGA LGG patients. At $\varepsilon = 10^{-2}$, the observed GPU discrepancy is within the requested threshold for 5/5 GSE24080 and 5/5 TCGA LGG patients. At $\varepsilon = 10^{-3}$, the observed GPU discrepancy is within the requested threshold for 4/5 GSE24080 and 2/5 TCGA LGG patients.

Agreement with the GPU degree-exact result alone does not establish exact numerical accuracy, because that baseline also contains FP32 rounding error. We therefore additionally compare both the GPU approximation and a separate FP64 CPU evaluation at the same selected nodes with an independent, tightly bounded 64-node FP64 reference. This reference has analytical quadrature bounds below 10^{-10} ; a 96-node cross-check is also available for the first patient of each cohort. It is a high-accuracy numerical reference, rather than a full degree-exact FP64 computation. The FP64 CPU evaluation is an accuracy diagnostic; its errors are never paired with the GPU timings.

The separate FP64 evaluation meets its requested threshold in 30/30 patient–tolerance cases, with a worst absolute difference of 1.59×10^{-10} , whereas the timed FP32 GPU evaluations meet

Dataset	Tolerance	GPU FP32		CPU FP64	
		Worst error	Pass	Worst error	Pass
GSE24080	$\varepsilon = 10^{-1}$	2.03×10^{-3}	5/5	1.21×10^{-11}	5/5
GSE24080	$\varepsilon = 10^{-2}$	2.15×10^{-3}	5/5	1.28×10^{-11}	5/5
GSE24080	$\varepsilon = 10^{-3}$	2.15×10^{-3}	1/5	1.28×10^{-11}	5/5
TCGA LGG	$\varepsilon = 10^{-1}$	3.05×10^{-2}	5/5	1.40×10^{-10}	5/5
TCGA LGG	$\varepsilon = 10^{-2}$	3.52×10^{-2}	3/5	1.31×10^{-10}	5/5
TCGA LGG	$\varepsilon = 10^{-3}$	3.91×10^{-2}	0/5	1.59×10^{-10}	5/5

Table 2: Separate precision checks against the same independent FP64 reference. Worst error is the maximum absolute feature-wise error over all five patients; Pass counts patients whose maximum error is at most their requested tolerance. CPU FP64 entries are accuracy diagnostics and are not the errors of the GPU runs timed in Table 1.

it in 19/30 cases against that same reference. For TCGA LGG methylation at $\varepsilon = 10^{-2}$, all five GPU approximations agree with their GPU degree-exact outputs within the threshold, but only 3/5 agree with the independent FP64 reference within that threshold (worst error 3.52×10^{-2}). This illustrates why agreement between two FP32 computations alone cannot establish the requested numerical accuracy. These comparisons distinguish certified quadrature approximation from end-to-end floating-point accuracy. Higher precision can reduce numerical error, but the analytical certificate itself does not include rounding. It also does not bound uncertainty from replacing a population background distribution with the chosen 30-row empirical background.

References

- [1] NCBI Gene Expression Omnibus. GSE24080: MAQC-II project: Multiple myeloma data set. <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE24080>, 2010. Accessed 19 September 2026.
- [2] UCSC Xena. TCGA brain lower grade glioma (LGG) DNA methylation (HumanMethylation450). <https://xenabrowser.net/datapages/?dataset=TCGA.LGG.sampleMap%2FHumanMethylation450&host=https%3A%2F%2Ftcga.xenahubs.net>. Accessed 19 September 2026.